EDS 223: Homework Assignment 3

Investigating Potential West Coast Marine Aquaculture

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Introduction

This analysis is intended to determine which Exclusive Economic Zones (EEZ) on the West Coast of the U.S. are best suited to developing marine aquaculture for several species (including oysters and red rock shrimp). Locations suitable for species were determined by meeting conditions for the species' optimal sea surface temperature (SST) AND depth. In this analysis, a function was made to make the workflow generalizable to determine the optimal marine aquaculture location for any species along the west coast. The output of the function is a map of EEZ regions colored by the amount of suitable area in each EEZ region based on the species, its minimum/maximum SST, and its minimum/maximum depth.

Load Packages

```
library(tidyverse)
library(terra)
library(sf)
library(tmap)
```

Prepare Data

```
# Read sea surface temperature (SST) data from 2008-2012
sst_2008 <- rast('data/average_annual_sst_2008.tif')
sst_2009 <- rast('data/average_annual_sst_2009.tif')
sst_2010 <- rast('data/average_annual_sst_2010.tif')
sst_2011 <- rast('data/average_annual_sst_2011.tif')
sst_2012 <- rast('data/average_annual_sst_2012.tif')

# Read bathymetry data
depth <- rast('data/depth.tif')

# Read exclusive economic zones (EEZ) data
eez <- st_read('data/wc_regions_clean.shp')
```

```r
# Combine the SST(2008-2012) rasters into a raster stack (one layer per year)
sst <- c(sst_2008, sst_2009, sst_2010, sst_2011, sst_2012)
```

```r
# ----Checking if the SST data, EEZ data, and depth data have the same CRS----
# Spatial operations require matching CRS, so print a message if they match and
raise a warning if they do not match

# Compare SST and depth
if(st_crs(sst) == st_crs(depth)){
  print("sst and depth CRS match!")
} else {
  warning("sst and depth CRS do not match.")
}

# Compare SST and EEZ
if(st_crs(sst) == st_crs(eez)){
  print("sst and eez CRS match!")
} else {
  warning("sst and eez CRS do not match.")
}

# Compare EEZ and depth
if(st_crs(eez) == st_crs(depth)){
  print("eez and depth CRS match!")
} else {
  warning("eez and depth CRS do not match.")
}
```

```
[1] "sst and depth CRS match!"
[1] "sst and eez CRS match!"
[1] "eez and depth CRS match!"
```

Process Data

```r
# Mean SST from 2008-2012 in Kelvin (into one layer)
avg_sst_K <- mean(sst, na.rm = TRUE)

# Convert average from Kelvin to Celsius
avg_sst_C <- avg_sst_K - 273.15

# Crop depth to match SST raster so only study area is processed
```

```
crop_depth <- crop(depth, ext(avg_sst_C))

# Reproject the cropped depth to SST's CRS and align it to the SST grid
crop_depth <- project(crop_depth, avg_sst_C)

# Resample depth to the `sst` grid so both rasters share the same resolution,
origin, and extent
# `crop_depth` provides the bathymetry values to transfer
# `avg_sst_C` provides the grid, its temperature values are ignored
resampled_depth <- resample(
                           crop_depth,
                           avg_sst_C,
                           method = 'bilinear'
                           )

# Confirm the resampled depth grid matches the SST grid (all should print TRUE)
cat("After resampling to SST grid:\n")
cat("Resolutions match:", all(res(avg_sst_C) == res(resampled_depth)), "\n")
cat("Origins match:", all(round(origin(avg_sst_C), 8) ==
                           round(origin(resampled_depth), 8)), "\n")
cat("Extents match:", ext(avg_sst_C) == ext(resampled_depth), "\n")
```

```
After resampling to SST grid:
Resolutions match: TRUE
Origins match: TRUE
Extents match: TRUE
```

```
# Stack SST and depth as a final alignment check:
# c() only works if the rasters share the same grid (will error if don't match)
c(avg_sst_C, resampled_depth)
```

Create a function to look at the suitable areas within West Coast EEZs for any species

```
# Function that maps the total suitable area for a species within each West
Coast EEZ
suitable_areas    <-    function(minimum_sst,    maximum_sst,    minimum_depth,
maximum_depth, species_name) {
    # Stop if the species' ranges are entered in the wrong order
    if (minimum_sst >= maximum_sst) {
    stop("minimum_sst must be less than maximum_sst.")
```

```r
}

if (minimum_depth >= maximum_depth) {
stop("minimum_depth must be less than maximum_depth.")
}

# ----Reclassify SST & depth values into suitable (1) and unsuitable (0)----

# Reclassification matrix for SST:
# each row is (from, to, new value)
# below the minimum_sst is unsuitable
# between the minimum_sst and maximum_sst is suitable
# above the maximum_sst is unsuitable
sst_rcl <- matrix(c(-Inf, minimum_sst, 0,
                    minimum_sst, maximum_sst, 1,
                    maximum_sst, Inf, 0),
                  ncol = 3, byrow = TRUE)

# Reclassification matrix for depth:
# below the minimum_depth is unsuitable
# between the minimum_depth and maximum_depth is suitable
# above the maximum_depth is unsuitable
depth_rcl <- matrix(c(-Inf, minimum_depth, 0,
                      minimum_depth, maximum_depth, 1,
                      maximum_depth, Inf, 0),
                    ncol = 3, byrow = TRUE)

# Apply the reclassification matrices

# New raster of 0s and 1s showing where SST is suitable
sst_suitable   <- classify(avg_sst_C, rcl = sst_rcl)
# New raster of 0s and 1s showing where depth is suitable
depth_suitable <- classify(resampled_depth, rcl = depth_rcl)

# ----Find locations that satisfy the species SST and depth requirements----

# Stack the SST and depth suitability rasters into one two-layer raster
sst_depth_zones <- c(sst_suitable, depth_suitable)

# Multiply the layers cell by cell: 1 * 1 = 1 only where both SST and depth
# are suitable (any 0 makes the result 0)
suitable_loc <- lapp(sst_depth_zones, fun = function(sst, depth) sst * depth)
```

```r
# ----Determine the most suitable EEZ----

# Set unsuitable cells (0) to NA so only suitable cells remain
suitable_loc[suitable_loc == 0] <- NA

# Rasterize the EEZ polygons onto the suitability grid so each cell holds
# its EEZ region name
eez_rast <- rasterize(eez, suitable_loc, field = "rgn")

# Keep only suitable cells that fall inside an EEZ
suitable_eez <- mask(suitable_loc, eez_rast)

# ----Find the area of grid cells----

# Calculate the area of each cell in km^2
# Cell area varies with latitude so it cannot be determined from
# resolution alone
cell_area <- cellSize(suitable_eez, unit = "km")

# Area of each suitable cell (unsuitable cells stay NA)
suitable_area <- suitable_eez * cell_area

# ----Find the total suitable area within each EEZ----

# Sum the suitable area within each EEZ zone
eez_suitable_area <- zonal(suitable_area,
                           eez_rast,
                           fun = "sum",
                           na.rm = TRUE) |>
                             # rename `lyr1` column that lists total
                             # suitable area within each EEZ to `area`
                             rename(area = lyr1)

# Create a tibble of `eez_suitable_area` to use for joining
zones_tibble <- as_tibble(eez_suitable_area)

# Join each EEZ's total suitable area to the EEZ polygons by region name
eez_zones <- left_join(eez, zones_tibble, by = "rgn")

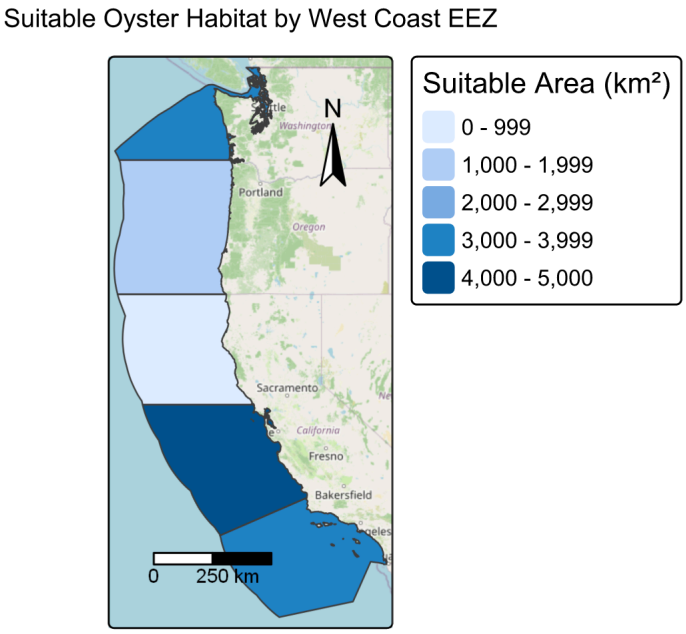

# ----Map the total suitable area in each EEZ----
tm_shape(eez_zones) +
    tm_basemap("OpenStreetMap") +
```

```
        tm_fill(
        fill = "area",
        fill.legend = tm_legend(title = "Suitable Area (km²)")
        ) +
        tm_scalebar(position = c("left", "bottom"),
                    breaks = c(0, 250, 500),
                    text.size = 0.6) +
        tm_compass(position = c("right", "top")) +
        tmap_options(component.autoscale = FALSE) +
        tm_borders(lwd = .8) +
        tm_title(paste("Suitable", species_name, "Habitat by West Coast EEZ"),
                 size = 0.80)

}
```

Using the function to look at Oysters

```
suitable_areas(minimum_sst = 11, maximum_sst = 30, minimum_depth = -70,
maximum_depth = 0, species_name = "Oyster")
```



Interpretation of the map

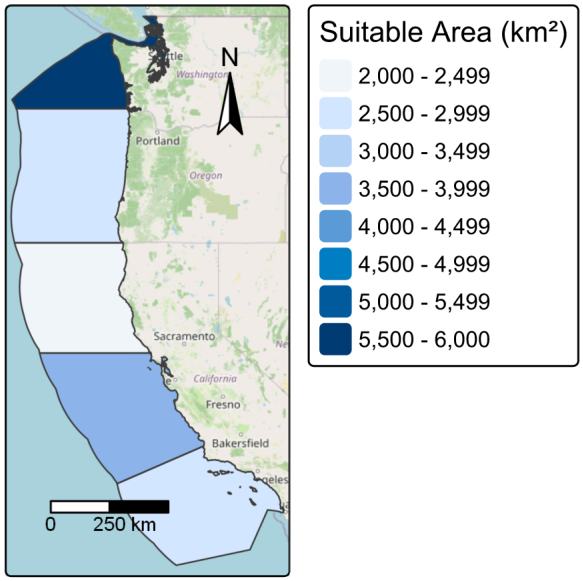
The optimal growth conditions for oysters are between 11 degrees Celsius and 30 degrees Celsius and between sea level and 70 m below sea level. The map featured above shows the Central California EEZ has the most amount of suitable area for oyster growth (4,374 km²). On the other

hand, the Northern California EEZ has the least amount of suitable area for oyster growth (324 km²).

Using the function to look at Red Rock Shrimp

```
suitable_areas(minimum_sst = 10, maximum_sst = 20, minimum_depth = -60,
maximum_depth = 0, species_name = "Red Rock Shrimp")
```

Suitable Red Rock Shrimp Habitat by West Coast EEZ



Interpretation of the map

The optimal growth conditions for red rock shrimp are between 10 degrees Celsius and 20 degrees Celsius and between sea level and 60 m below sea level. The map featured above shows the Washington EEZ has the most amount of suitable area for red rock shrimp growth (5,630 km²). On the other hand, the Northern California EEZ has the least amount of suitable area for red rock shrimp growth (2,311 km²). Northern California ranked lowest for both species, which suggests that conditions in this zone may limit aquaculture potential in general.

Data Citations:

- Flanders Marine Institute (2023). Maritime Boundaries Geodatabase: Maritime Boundaries and Exclusive Economic Zones (200NM), version 12. Accessed on November 02, 2026 at https://www.marineregions.org/eez.php.
- GEBCO Bathymetric Compilation Group 2026 (2026). The GEBCO_2026 Grid - a continuous terrain model for oceans and land at 15 arc-second intervals. Accessed on November 02, 2026 at https://www.gebco.net/data-products/gridded-bathymetry-data#area.

- NOAA Coral Reef Watch (2026). NOAA Coral Reef Watch Version 3.1 Daily Global 5km Satellite Sea Surface Temperature Anomaly Product, 2008–2012. College Park, Maryland, USA: NOAA Coral Reef Watch. Accessed on November 02, 2026 at https://coralreefwatch.noaa.gov/product/5km/index_5km_ssta.php.